

El Niño and the Ski Season

What the Oceanic Niño Index actually predicts about powder days, storm frequency and base depth across 37 ski regions of the western United States — and, just as usefully, where it predicts nothing at all.

932 SNOTEL STATIONS	40.2M DAILY OBSERVATIONS	1964–2026 WATER YEARS
37 SKI REGIONS	806 TESTS RUN	31 SURVIVE FDR

Prepared 20 September 2026 · Data: NOAA CPC · NRCS AWDB · NCEI nClimDiv · Method after Brown & Harper (2026), HESS 30, 5735

THE SHORT VERSION

Four findings

1. The effect is real but geographically narrow. Of 806 tests across every region, window of winter and measure, 31 survive false-discovery control. Nearly all of them sit in the northern Rockies and the Cascades, where El Niño costs roughly a foot of snow, or in New Mexico and Arizona, where it adds snow in spring.

2. For most of the West, El Niño predicts little, and the standard of proof matters. Under correction across the whole 806 tests, the Wasatch, the I-70 corridor, Tahoe, the San Juans, Sun Valley and the Elk Mountains never clear significance on any measure in any window. Judged one region at a time, several of them do separate from zero over the holidays. Both statements are in this report; the difference between them is the price of having asked the question many times.

3. The Christmas–New Year window behaves differently from the rest of winter. Every other window follows a north–south seesaw. The holidays lean toward less snow in 85 of 102 tests, in all three latitude bands at once, including the southern ranges that otherwise gain.

4. The strength of an event carries no information. Tested directly across 2158 region-window-measure combinations, nothing survives. A very strong El Niño delivers more powder days than a strong one.

El Niño brings less snow

El Niño brings more snow

SURVIVES FDR

 clears false-discovery control across the whole grid

RESULT

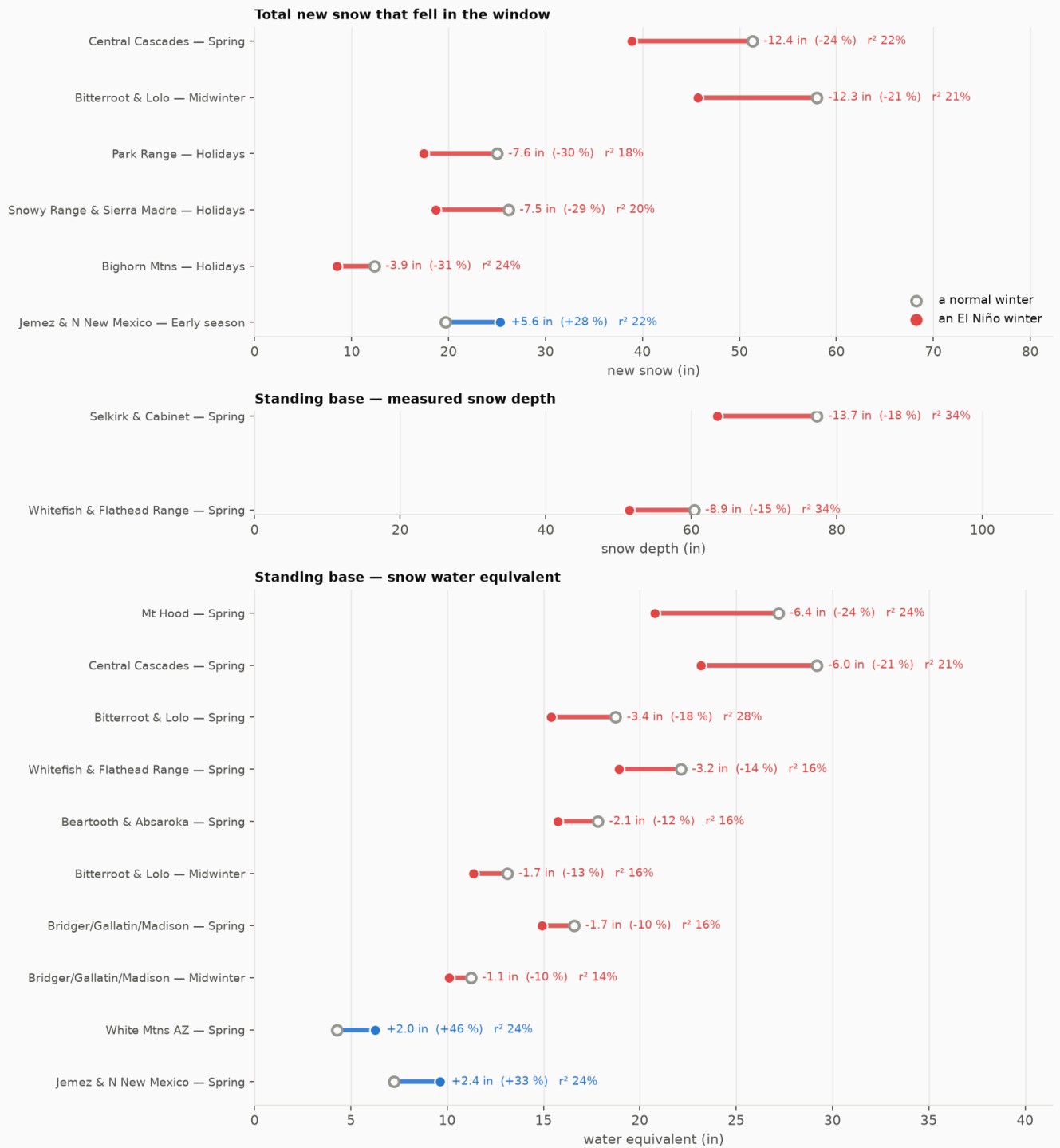
What survives, in inches

These are the region–window–measure combinations that clear Benjamini–Hochberg false-discovery control across all 806 tests. Each row compares the long-run average of every winter on record against the average of El Niño winters only.

Inches. New snow is what fell during the window; base is what was standing, measured both as depth and as water equivalent.								
SKI REGION	WINDOW	MEASURE	WINTERS	NORMAL	EL NIÑO	DIFFERENCE	CHANGE	R²
Bighorn Mtns	Holidays	Total new snow	48	12.3	8.5	-3.9 in	-31%	24%
Park Range (Steamboat)	Holidays	Total new snow	47	25.0	17.4	-7.6 in	-30%	18%
Snowy Range & Sierra Madre	Holidays	Total new snow	46	26.2	18.7	-7.5 in	-29%	20%
Central Cascades (Stevens/Snoqualmie)	Spring	Total new snow	45	51.3	38.9	-12.4 in	-24%	22%
Mt Hood	Spring	Base, water equivalent	48	27.2	20.8	-6.4 in	-24%	24%
Bitterroot & Lolo (Montana Snowbowl)	Midwinter	Total new snow	59	58.0	45.7	-12.3 in	-21%	21%

Ski Region	Window	Measure	Winters	Normal	El Niño	Difference	Change	R²
Central Cascades (Stevens/Snoqualmie)	Spring	Base, water equivalent	45	29.2	23.2	-6.0 in	-21%	21%
Bitterroot & Lolo (Montana Snowbowl)	Spring	Base, water equivalent	59	18.7	15.4	-3.4 in	-18%	28%
Selkirk & Cabinet (Schweitzer/Silver)	Spring	Base, snow depth	25	77.3	63.5	-13.7 in	-18%	34%
Whitefish & Flathead Range	Spring	Base, snow depth	25	60.4	51.5	-8.9 in	-15%	34%
Whitefish & Flathead Range	Spring	Base, water equivalent	52	22.1	18.9	-3.2 in	-14%	16%
Bitterroot & Lolo (Montana Snowbowl)	Midwinter	Base, water equivalent	59	13.1	11.4	-1.7 in	-13%	16%
Beartooth & Absaroka (Red Lodge)	Spring	Base, water equivalent	53	17.8	15.7	-2.1 in	-12%	16%
Bridger/Gallatin/Madison (Big Sky)	Spring	Base, water equivalent	60	16.6	14.9	-1.7 in	-10%	16%
Bridger/Gallatin/Madison (Big Sky)	Midwinter	Base, water equivalent	60	11.2	10.1	-1.1 in	-10%	14%
Jemez & N New Mexico	Early season	Total new snow	46	19.7	25.3	+5.6 in	+28%	22%
Jemez & N New Mexico	Spring	Base, water equivalent	46	7.2	9.6	+2.4 in	+33%	24%
White Mtns AZ (Sunrise Park)	Spring	Base, water equivalent	46	4.3	6.3	+2.0 in	+46%	24%

What El Niño changes, in inches — only results that survive false-discovery control



Only the results that survive false-discovery control, in inches. Three separate axes: inches of falling snow, inches of standing snow depth and inches of water in that snow are different quantities and cannot share a scale. The open circle is a normal winter; the filled circle is an El Niño winter.

RESULT

What survives, in days

Days within the window. A powder day is a day with at least six inches of new snow; a storm day at least two.								
SKI REGION	WINDOW	MEASURE	WINTERS	NORMAL	EL NIÑO	DIFFERENCE	CHANGE	R²
Bitterroot & Lolo (Montana Snowbowl)	Midwinter	Powder days (≥6 in)	59	1.5	0.8	-0.7 d	-45%	20%
Bighorn Mtns	Holidays	Storm days (≥2 in)	48	1.8	1.1	-0.7 d	-39%	19%
Park Range (Steamboat)	Holidays	Storm days (≥2 in)	47	4.3	2.7	-1.5 d	-36%	23%
Snowy Range & Sierra Madre	Holidays	Storm days (≥2 in)	46	4.3	3.0	-1.3 d	-30%	21%
Central Cascades (Stevens/Snoqualmie)	Spring	Storm days (≥2 in)	45	8.7	6.4	-2.2 d	-26%	26%
Bitterroot & Lolo (Montana Snowbowl)	Midwinter	Storm days (≥2 in)	59	9.2	7.1	-2.1 d	-23%	18%
Bridger/Gallatin/Madison (Big Sky)	Midwinter	Days with a skiable base	60	29.9	25.1	-4.8 d	-16%	13%
Bitterroot & Lolo (Montana Snowbowl)	Spring	Days with a skiable base	59	30.3	25.8	-4.5 d	-15%	24%
Bitterroot & Lolo (Montana Snowbowl)	Midwinter	Days with a skiable base	59	26.3	22.8	-3.5 d	-13%	15%
Whitefish & Flathead Range	Spring	Days with a skiable base	52	38.0	33.0	-4.9 d	-13%	18%
Jemez & N New Mexico	Early season	Storm days (≥2 in)	46	3.4	4.3	+0.9 d	+28%	20%
Jemez & N New Mexico	Spring	Days with a skiable base	46	15.0	23.8	+8.8 d	+58%	20%
White Mtns AZ (Sunrise Park)	Spring	Days with a skiable base	46	8.1	14.1	+6.0 d	+74%	18%



READING THE LARGEST EFFECTS

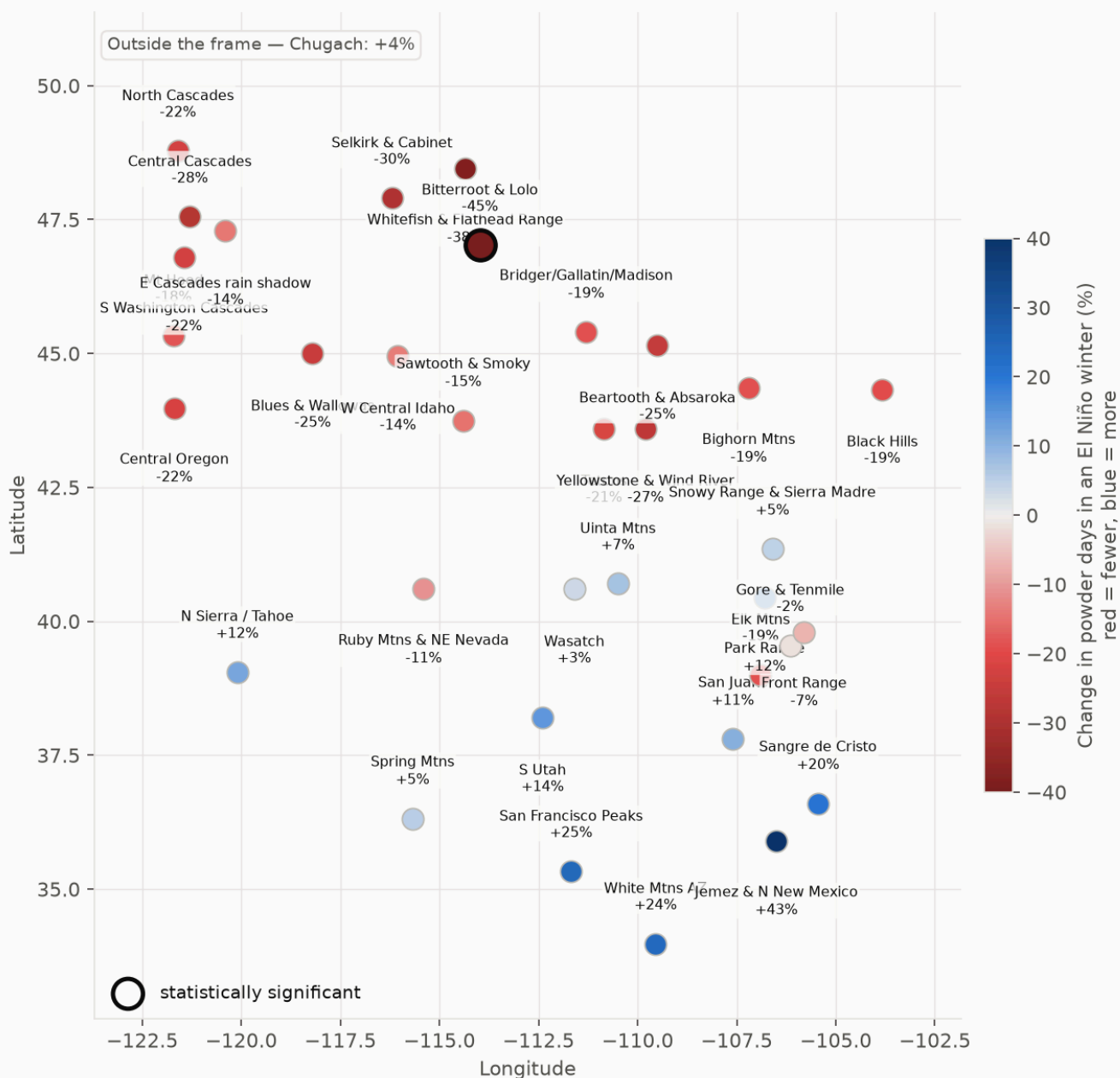
Montana Snowbowl in midwinter is the single most affected place in the study: twelve inches less new snow, two fewer storm days, and 45 % fewer powder days. The largest gain anywhere is northern New Mexico in spring, which picks up nearly nine additional days with a skiable base.

GEOGRAPHY

Where the effect lives

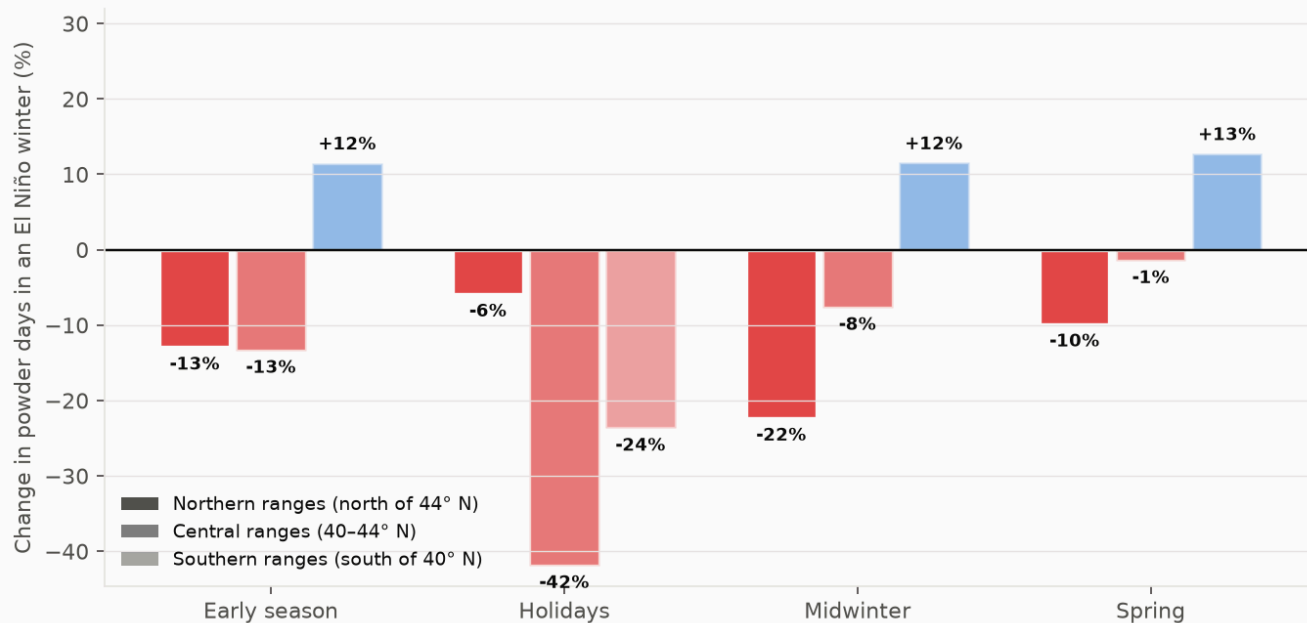
The turnover sits near 40–42° N, running through central Colorado and Utah. North of it El Niño suppresses snowfall; south of it, in spring, it adds. The southern half of the seesaw is weaker and less reliable than the northern half — several southern regions show a positive tendency that never reaches significance.

Where El Niño costs powder days — midwinter



Change in midwinter powder days by region. Ringed markers survive false-discovery control.

El Niño's effect on powder days, by window of winter and latitude

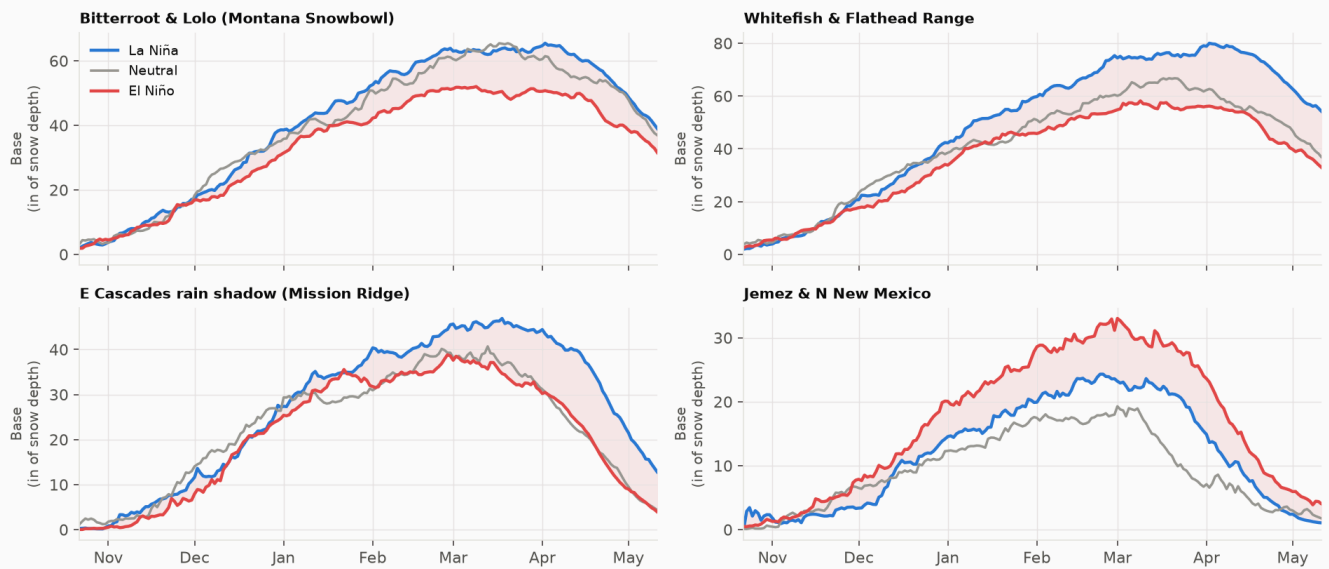


Change in powder days by window of winter, split at the ENSO node. Averaging all regions together would cancel the seesaw to near zero and say nothing true about anywhere; splitting makes both halves visible — and exposes the holiday window, which is negative in all three bands at once.

Direction consistency by window. Each test is one region and one measure; the sign test asks whether the direction holds across independent regions, which is the question that matters when each individual correlation is modest.

WINDOW OF WINTER	TESTS	LEANING TO LESS SNOW	SHARE	MEAN R	SIGN-TEST P
Early season	102	71 of 102	70%	-0.058	9.3e-05
Holidays	102	85 of 102	83%	-0.132	4.7e-12
Midwinter	102	76 of 102	75%	-0.107	7.4e-07
Spring	102	65 of 102	64%	-0.066	7.2e-03

How the base builds through the winter, by ENSO phase worst-hit regions at the top, a region that gains at the bottom right



Base depth through the winter by ENSO phase, in inches of measured snow depth. Two worst-hit regions at the top, a near-neutral region and a gaining region below.

GEOGRAPHY WITH UNCERTAINTY

The effect moves through the winter

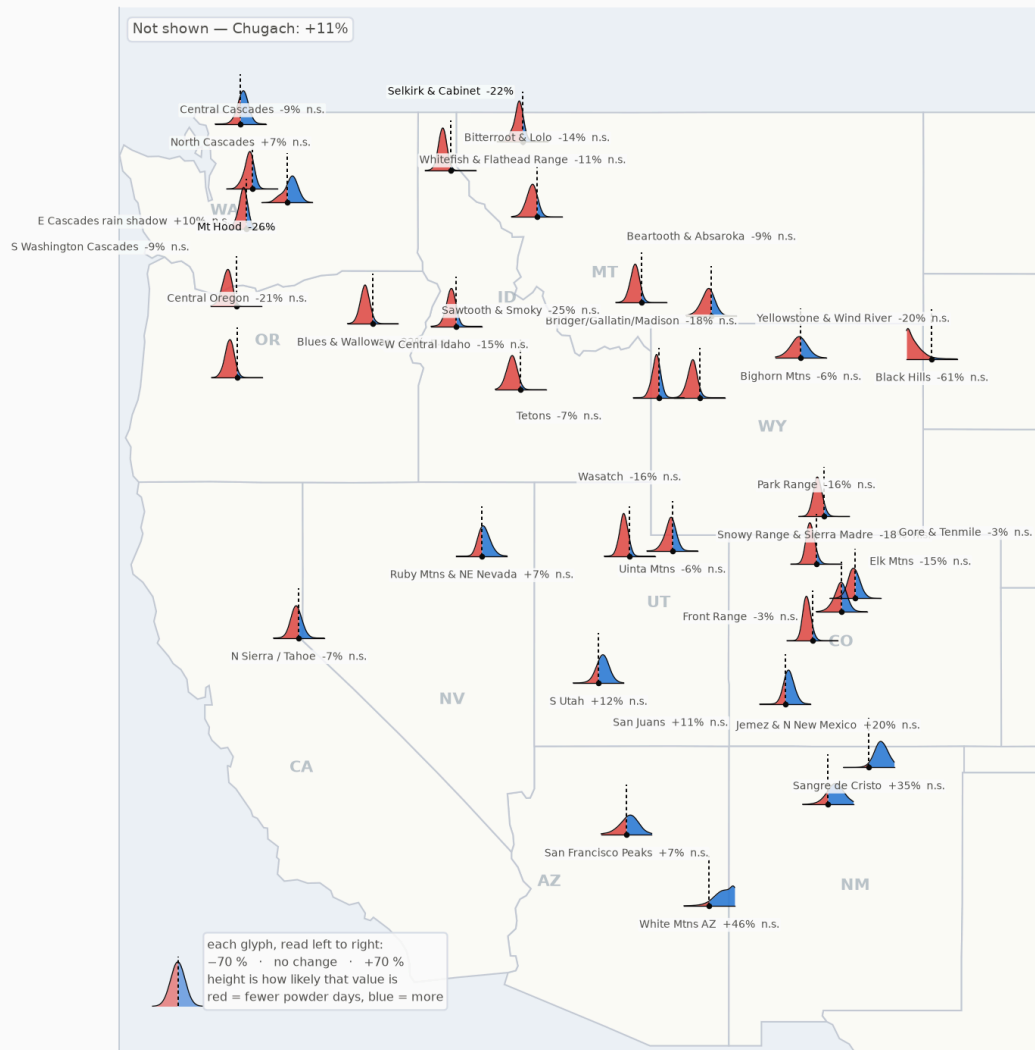
A single number per region hides how firmly it is pinned, so every ski region here carries its whole probability distribution rather than a colour. The dashed hairline through each glyph is no change. A density sitting astride that line is a region where El Niño does nothing measurable.

Drawn window by window, the maps show something a season-long average cannot: **the effect does not sit still.** It is almost absent in the early season, seizes the interior ranges over the holidays, shifts to the northern Rockies and Cascades in midwinter, and retreats to the Cascades alone by spring. A region can be firmly hit in one window and indistinguishable from zero in the next.

Early season · 1 November – 15 December

Almost nothing is decided. Only Mt Hood and the Selkirks separate from zero. Whatever El Niño does to a season, it has barely started by mid-December.

The change in powder days in every ski region, with its uncertainty — early season



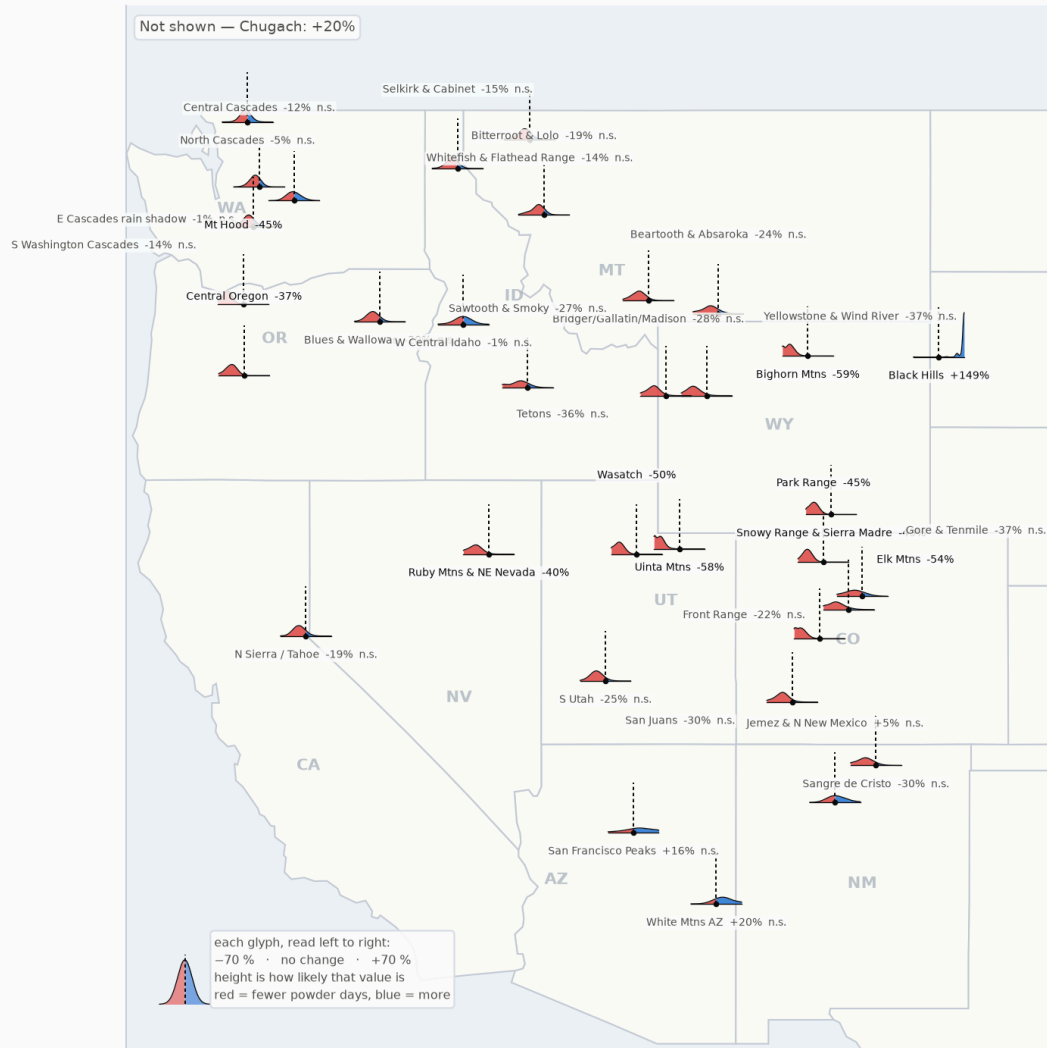
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 31 of 33 regions straddle zero and are marked n.s.

Two of thirty-four regions clear zero. The rest are undecided.

Holidays · 16 December – 5 January

The widest, most coherent signal of the winter, and it lands on the interior: the Wasatch, the Uintas, the Elk Mountains, the Park and Snowy ranges, the Rubies. These are the same regions that show nothing at all in midwinter. The Wasatch loses eight of the twenty-three inches it normally gets over Christmas and New Year.

The change in powder days in every ski region, with its uncertainty — holidays



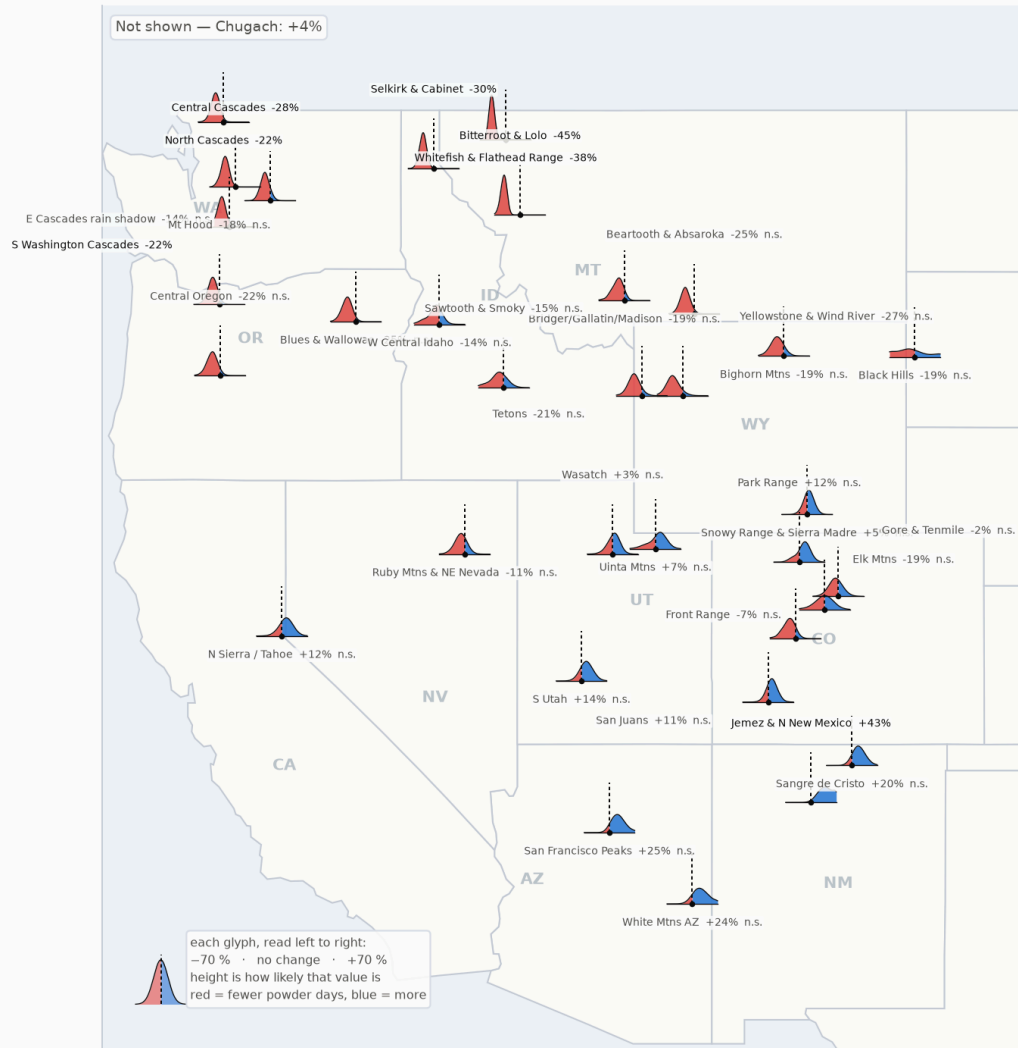
Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 23 of 33 regions straddle zero and are marked n.s.

Ten of thirty-four regions clear zero, nine of them losses, and the cluster sits over the Great Basin and the Colorado and Wyoming ranges rather than the north.

Midwinter • 6 January – 28 February

The signal migrates north. The Bitterroots, Whitefish, the Selkirks and all three Washington Cascade regions separate from zero, and northern New Mexico does so in the other direction. The interior ranges that were firmly hit over the holidays are now undecided.

The change in powder days in every ski region, with its uncertainty — midwinter

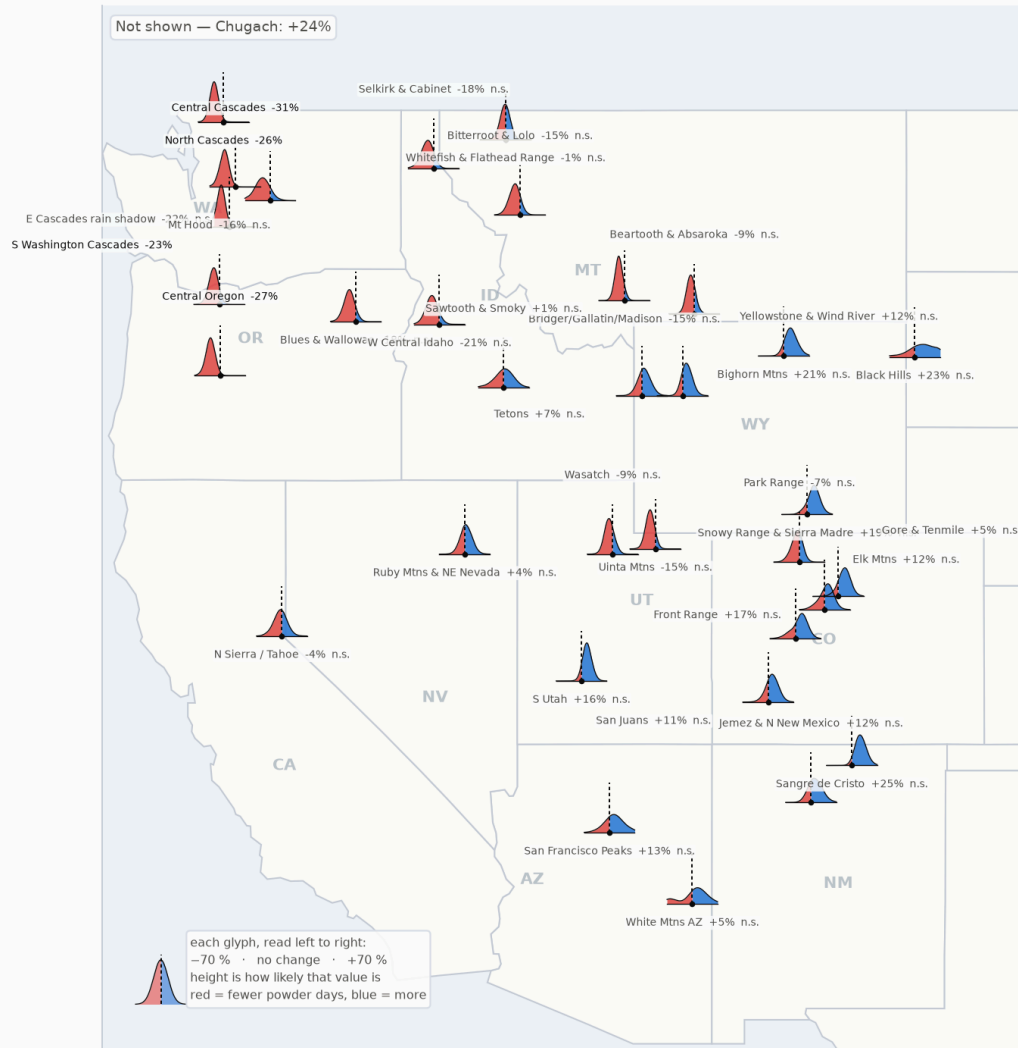


Seven of thirty-four regions clear zero: six northern losses and one southern gain.

Spring · 1 March – 15 April

Only the Cascades and central Oregon remain decided. Everywhere else the distributions have widened back across zero.

The change in powder days in every ski region, with its uncertainty — spring



Each glyph is a probability density from 10,000 resamples of the record, calibrated so its width is an honest sampling distribution. 29 of 33 regions straddle zero and are marked n.s.

Four of thirty-four regions clear zero, all of them in the Cascades and Oregon.

TWO STANDARDS OF EVIDENCE, AND WHY THEY DISAGREE

These maps judge each region on its own: does its 95 % interval exclude zero? The headline tables earlier apply Benjamini–Hochberg correction across all 806 tests at once, which is far stricter, because asking a question of 34 regions in 4 windows on 6 measures will turn up apparent effects by chance alone.

That is why the Wasatch appears here at -50 % over the holidays and does not appear in the list of results that survive correction. The per-region interval is the right tool for asking "what does the record say about *this* range"; the corrected grid is the right tool for asking "which of these findings would I bet on". A region that clears the first but not the second is worth watching, not worth relying on.

Every region whose 95 % interval excludes zero, by window. Percentages of powder days are shown next to the inches of new snow behind them, because a window of twenty-one days has few powder days anywhere and a large percentage can rest on

a fraction of a day.								
WINDOW	SKI REGION	POWDER DAYS, NORMAL	CHANGE	NEW SNOW, NORMAL	EL NIÑO	DIFFERENCE	95 % INTERVAL	
Early season	Mt Hood	2.29	-26%	51 in	42 in	-10 in	-54% to -3%	
	Selkirk & Cabinet	2.01	-22%	55 in	50 in	-5 in	-45% to -0%	
Holidays	Bighorn Mtns TINY BASE	0.13	-59%	12 in	8 in	-4 in	-115% to -23%	
	Uinta Mtns	0.54	-58%	17 in	11 in	-6 in	-91% to -30%	
	Elk Mtns	0.59	-54%	21 in	15 in	-6 in	-88% to -22%	
	Wasatch	0.97	-50%	23 in	15 in	-8 in	-81% to -22%	
	Snowy Range & Sierra Madre	0.85	-49%	26 in	19 in	-8 in	-78% to -21%	
	Park Range	0.85	-45%	25 in	17 in	-8 in	-72% to -19%	
	Mt Hood	1.53	-45%	33 in	26 in	-7 in	-72% to -19%	
	Ruby Mtns & NE Nevada	0.65	-40%	19 in	14 in	-5 in	-82% to -3%	
	Central Oregon	1.56	-37%	35 in	28 in	-6 in	-71% to -8%	
	Black Hills TINY BASE	0.11	+149%	10 in	11 in	+1 in	+50% to +355%	
Midwinter	Bitterroot & Lolo	1.52	-45%	58 in	46 in	-12 in	-62% to -29%	
	Whitefish & Flathead Range	1.54	-38%	59 in	48 in	-11 in	-53% to -24%	
	Selkirk & Cabinet	2.53	-30%	71 in	60 in	-11 in	-49% to -11%	
	Central Cascades	3.70	-28%	85 in	71 in	-14 in	-52% to -7%	
	North Cascades	4.59	-22%	102 in	90 in	-13 in	-47% to -1%	
	S Washington Cascades	3.93	-22%	89 in	79 in	-10 in	-47% to -1%	
	Jemez & N New Mexico	0.73	+43%	30 in	38 in	+8 in	+5% to +90%	
Spring	Central Cascades	1.74	-31%	51 in	39 in	-12 in	-57% to -7%	
	Central Oregon	1.90	-27%	54 in	46 in	-9 in	-52% to -3%	
	North Cascades	2.69	-26%	69 in	60 in	-9 in	-49% to -3%	
	S Washington Cascades	2.09	-23%	61 in	50 in	-11 in	-46% to -2%	

ONE NUMBER NOT TO TRUST

The Black Hills reads +149 % over the holidays. That is 0.11 powder days becoming 0.27 — a sixth of a day, at the one region in the study with two SNOTEL stations. Its interval excludes zero, but the quantity is too small to carry meaning. The same caution applies more mildly to the Bighorns, where a 59 % loss is 0.13 days becoming 0.05. Read the inches column for these.

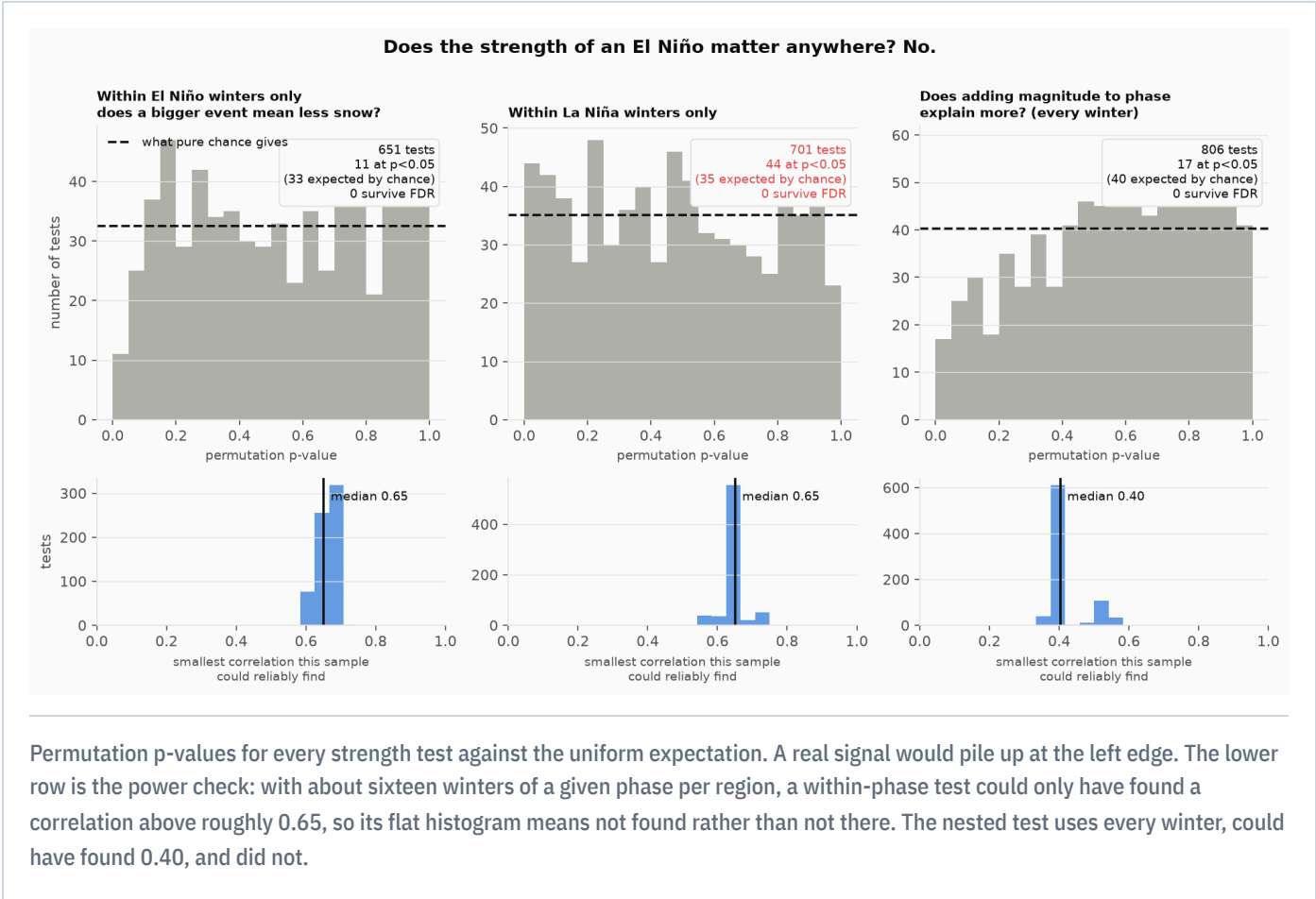
THE NEGATIVE RESULT

Event strength carries no information

Phase says which way a winter leans. Strength says how far the index went. The two are routinely conflated in seasonal forecasting, so strength was tested head-on rather than inferred: every region, window and measure was asked whether a larger event means less snow among El Niño winters only, the same among La Niña winters, and whether adding the index value to a phase-only model explains more.

2158 strength tests with permutation p-values and false-discovery control across the whole family.					
TEST	TESTS RUN	P < 0.05	EXPECTED BY CHANCE	SURVIVE FDR	SMALLEST R FINDABLE
Within El Niño winters only	651	11	33	0	0.65
Within La Niña winters only	701	44	35	0	0.65
Adding magnitude to phase (every winter)	806	17	40	0	0.40

Nothing survives. The p-values are distributed exactly as pure chance distributes them, and the within-El Niño family returns *fewer* nominal hits than chance alone would produce.



Midwinter powder days by strength bin, averaged over every region. If magnitude mattered the bars would descend from left to right.			
PHASE AND STRENGTH	WINTERS	POWDER DAYS	
La Niña · strong	6	1.59	
La Niña · moderate	5	1.30	
La Niña · weak	10	1.32	
Neutral · neutral	18	1.30	
El Niño · weak	9	1.16	
El Niño · moderate	4	1.15	
El Niño · strong	4	1.04	
El Niño · very strong	3	1.37	



Variance decomposition. Knowing only the phase beats knowing the index value; adding magnitude makes the prediction worse in every window.

Window	R ² From Phase Alone	R ² From The Index	Change From Adding Strength	Tests Where It Helps	R ² Within El Niño Only
Early season	6.1%	4.0%	-2.1%	18 of 198	4.6%
Holidays	7.4%	5.0%	-2.4%	8 of 201	7.0%
Midwinter	6.8%	5.4%	-1.3%	33 of 203	7.2%
Spring	9.7%	6.4%	-3.3%	28 of 204	7.2%

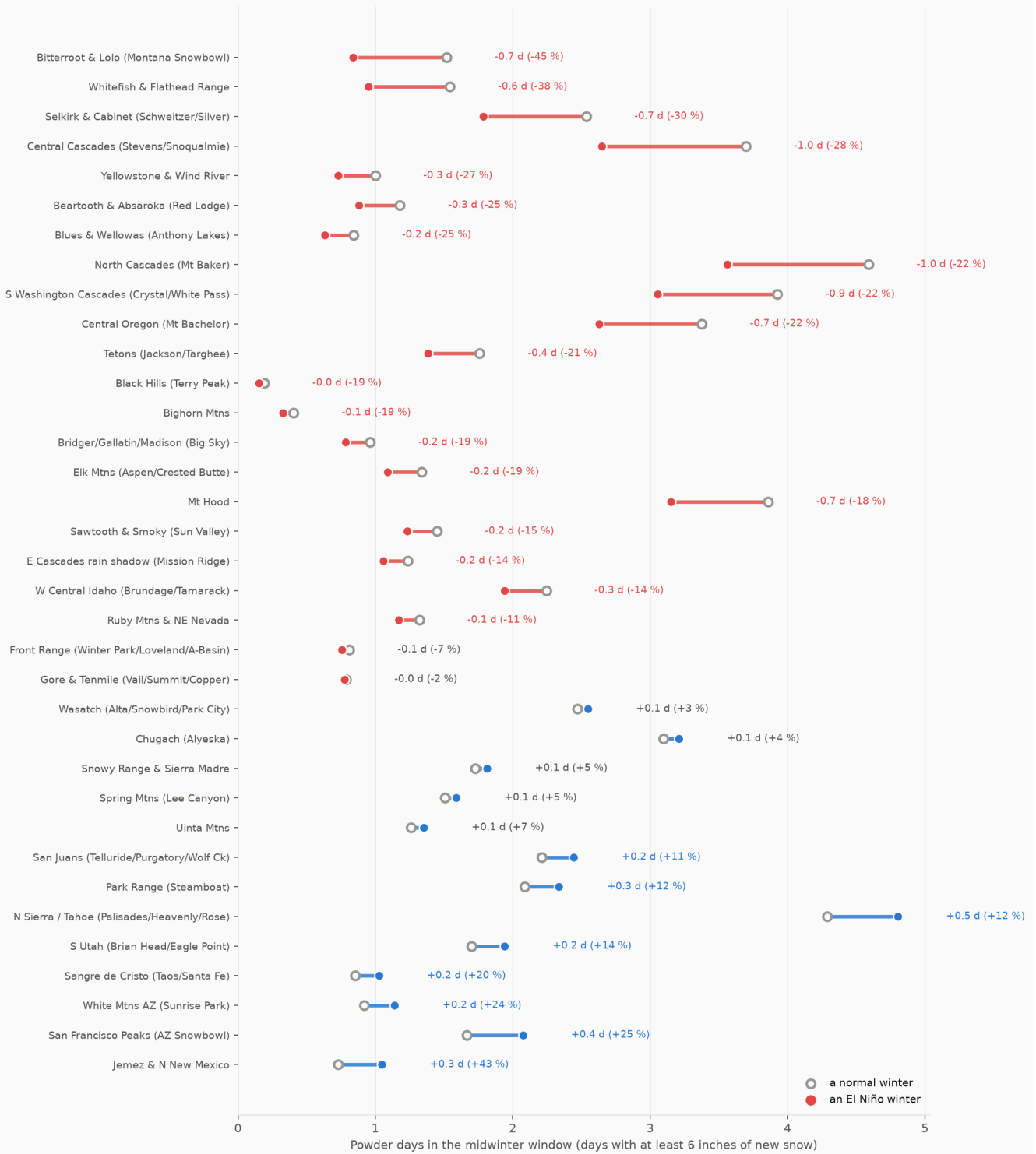
Plan for the phase and ignore the magnitude. The three very strong events on record (1983, 1998, 2016) were not the worst winters in the northern Rockies, and the strong events (1958, 1966, 1973, 1992, 2010, 2024) were not reliably worse than the weak ones.

The complete picture, including the regions where nothing is detectable. Ordered from the largest loss to the largest gain. Regions with under one powder day a month show large percentage swings on very small absolute differences – read the day counts, not the percentages, at the extremes.

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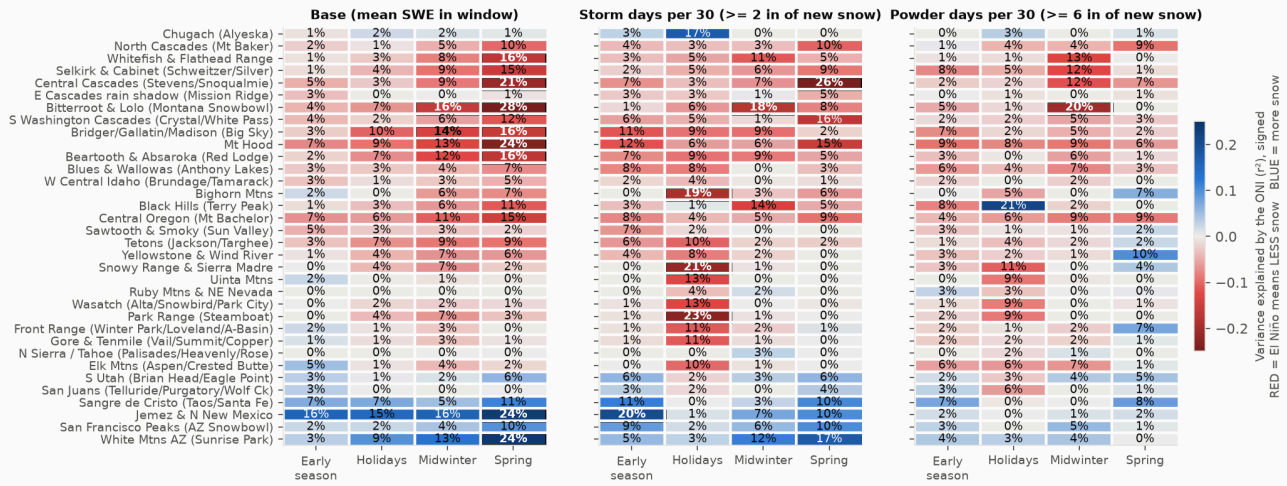
Ski Region	States	Snow Climate	Winters	Normal	El Niño	Diff	Change	La Niña	Status
Bitterroot & Lolo (Montana Snowbowl)	MT+ID	transitional	59	1.5	0.8	-0.7	-45%	2.0	SURVIVES FDR
Whitefish & Flathead Range	MT	transitional	52	1.5	1.0	-0.6	-38%	2.0	SURVIVES FDR
Selkirk & Cabinet (Schweitzer/Silver)	ID+MT	transitional	46	2.5	1.8	-0.7	-30%	3.2	SURVIVES FDR
Central Cascades (Stevens/Snoqualmie)	WA	maritime	45	3.7	2.6	-1.0	-28%	4.4	SURVIVES FDR
Yellowstone & Wind River	WY	continental	48	1.0	0.7	-0.3	-27%	0.8	NOT SIGNIFICANT
Beartooth & Absaroka (Red Lodge)	MT+WY	continental	53	1.2	0.9	-0.3	-25%	1.5	SURVIVES FDR
Blues & Wallowas (Anthony Lakes)	OR	transitional	48	0.8	0.6	-0.2	-25%	0.9	NOT SIGNIFICANT
North Cascades (Mt Baker)	WA	maritime	44	4.6	3.6	-1.0	-22%	5.1	NOT SIGNIFICANT
S Washington Cascades (Crystal/White Pass)	WA	maritime	45	3.9	3.1	-0.9	-22%	4.3	NOT SIGNIFICANT
Central Oregon (Mt Bachelor)	OR	transitional	46	3.4	2.6	-0.7	-22%	4.2	NOT SIGNIFICANT
Tetons (Jackson/Targhee)	WY+ID	intermountain	46	1.8	1.4	-0.4	-21%	1.6	NOT SIGNIFICANT
Black Hills (Terry Peak)	SD	continental	26	0.2	0.2	-0.0	-19%	0.3	NOT SIGNIFICANT
Bighorn Mtns	WY	continental	48	0.4	0.3	-0.1	-19%	0.5	SURVIVES FDR
Bridger/Gallatin/Madison (Big Sky)	MT	continental	60	1.0	0.8	-0.2	-19%	1.0	SURVIVES FDR
Elk Mtns (Aspen/Crested Butte)	CO	continental	46	1.3	1.1	-0.2	-19%	1.5	NOT SIGNIFICANT
Mt Hood	OR	maritime	48	3.9	3.1	-0.7	-18%	4.7	SURVIVES FDR
Sawtooth & Smoky (Sun Valley)	ID	intermountain	46	1.4	1.2	-0.2	-15%	1.3	NOT SIGNIFICANT
E Cascades rain shadow (Mission Ridge)	WA	transitional	45	1.2	1.1	-0.2	-14%	1.5	NOT SIGNIFICANT
W Central Idaho (Brundage/Tamarack)	ID	intermountain	46	2.2	1.9	-0.3	-14%	2.3	NOT SIGNIFICANT
Ruby Mtns & NE Nevada	NV	continental	46	1.3	1.2	-0.1	-11%	1.4	NOT SIGNIFICANT
Front Range (Winter Park/Loveland/A-Basin)	CO	continental	48	0.8	0.8	-0.1	-7%	0.8	NOT SIGNIFICANT
Gore & Tenmile (Vail/Summit/Copper)	CO	continental	48	0.8	0.8	-0.0	-2%	0.8	NOT SIGNIFICANT
Wasatch (Alta/Snowbird/Park City)	UT	intermountain	48	2.5	2.5	+0.1	+3%	2.4	NOT SIGNIFICANT
Chugach (Alyeska)	AK	maritime	43	3.1	3.2	+0.1	+4%	3.1	NOT SIGNIFICANT
Snowy Range & Sierra Madre	WY	continental	46	1.7	1.8	+0.1	+5%	1.7	SURVIVES FDR
Spring Mtns (Lee Canyon)	NV	continental	18	1.5	1.6	+0.1	+5%	1.3	NOT SIGNIFICANT
Uinta Mtns	UT	continental	48	1.3	1.4	+0.1	+7%	1.1	NOT SIGNIFICANT
San Juans (Telluride/Purgatory/Wolf Ck)	CO	continental	47	2.2	2.4	+0.2	+11%	2.2	NOT SIGNIFICANT
Park Range (Steamboat)	CO	continental	47	2.1	2.3	+0.3	+12%	2.1	SURVIVES FDR
N Sierra / Tahoe (Palisades/Heavenly/Rose)	CA+NV	maritime	48	4.3	4.8	+0.5	+12%	4.1	NOT SIGNIFICANT
S Utah (Brian Head/Eagle Point)	UT	continental	48	1.7	1.9	+0.2	+14%	1.5	NOT SIGNIFICANT
Sangre de Cristo (Taos/Santa Fe)	NM+CO	continental	47	0.9	1.0	+0.2	+20%	0.7	NOT SIGNIFICANT
White Mtns AZ (Sunrise Park)	AZ	continental	46	0.9	1.1	+0.2	+24%	0.6	SURVIVES FDR
San Francisco Peaks (AZ Snowbowl)	AZ	continental	46	1.7	2.1	+0.4	+25%	1.4	NOT SIGNIFICANT
Jemez & N New Mexico	NM	continental	46	0.7	1.0	+0.3	+43%	0.7	SURVIVES FDR

What El Niño costs in powder days — midwinter



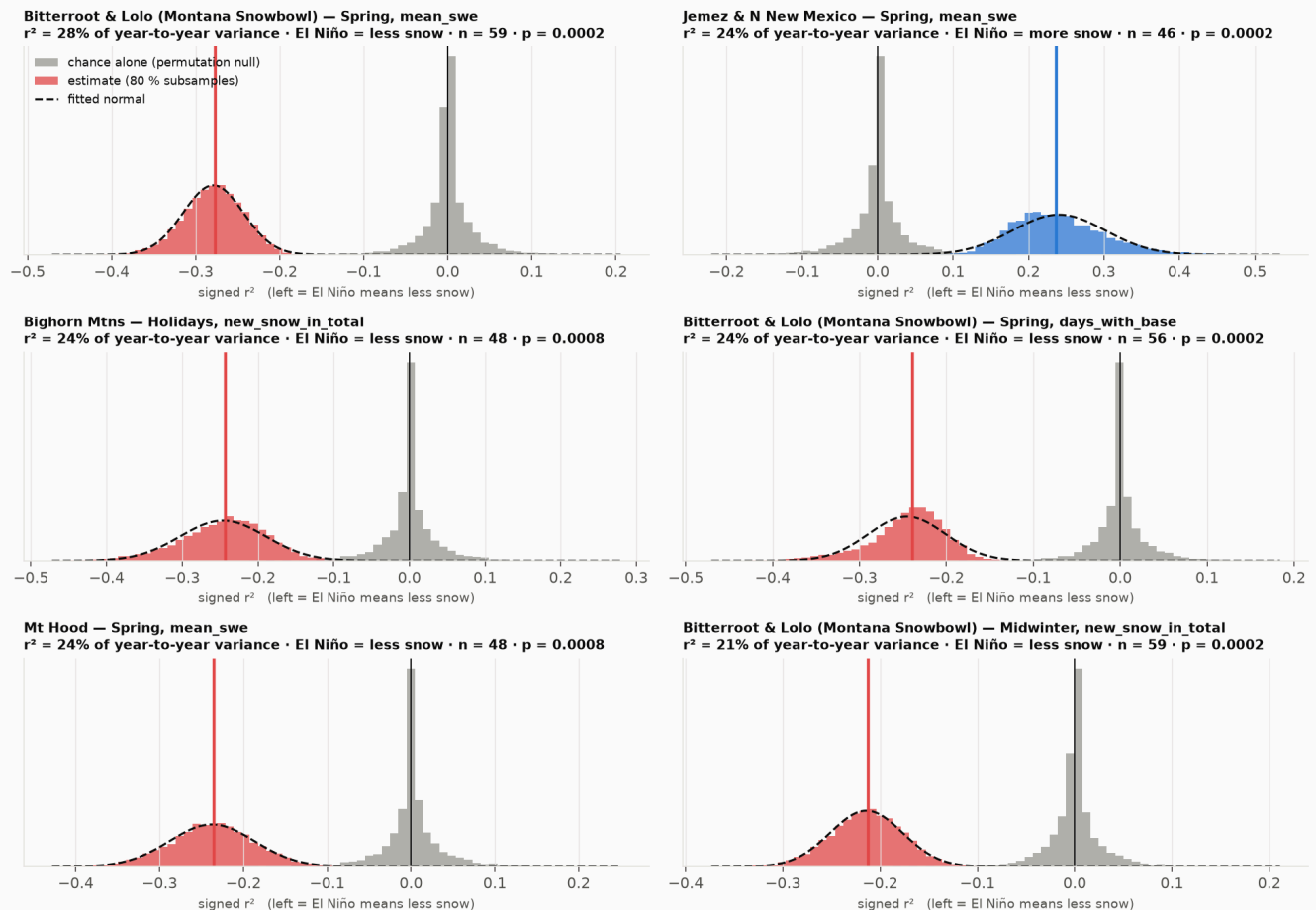
Powder days in a normal winter against an El Niño winter, every region with data, sorted by effect.

El Niño and the ski season, by region and window of winter

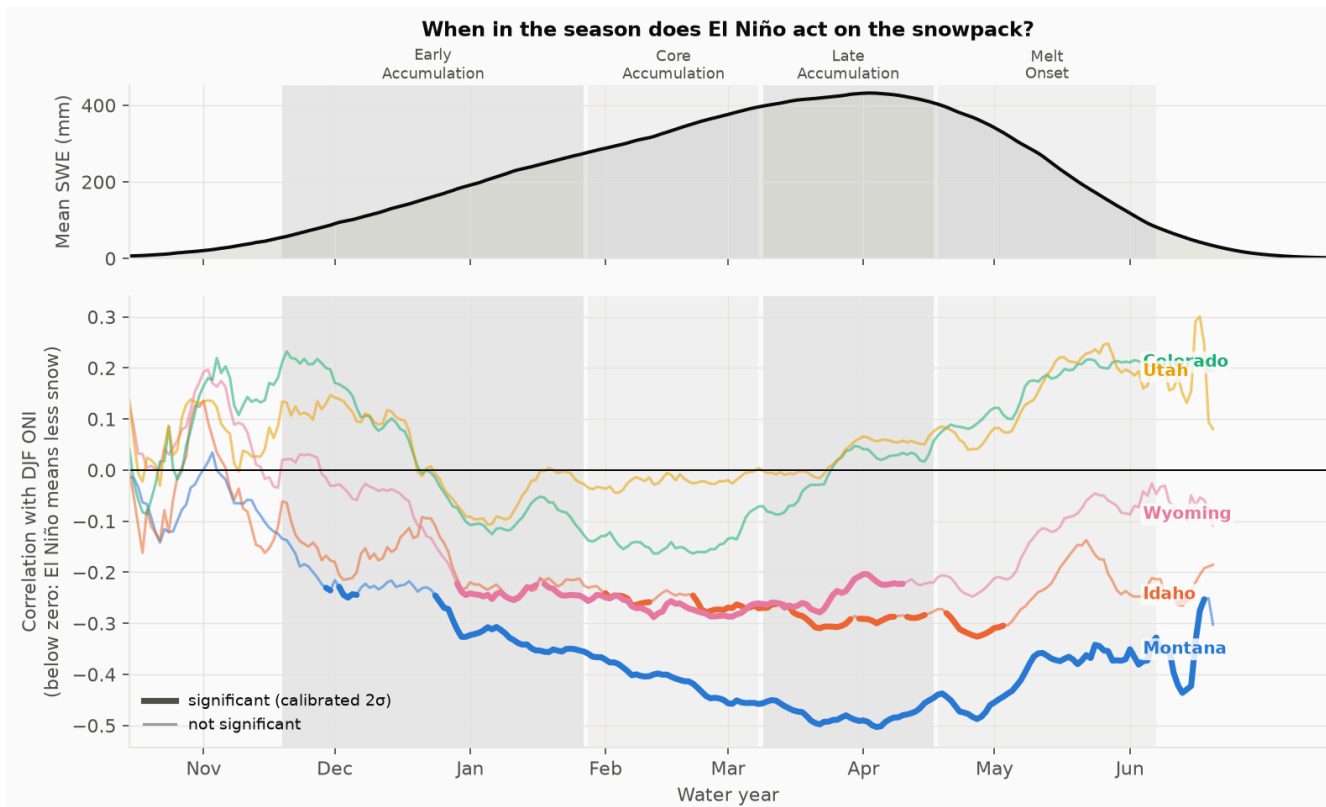


Variance explained (r^2) by region, window and measure, regions ordered north to south. The vertical gradient is the seesaw.

Probability distributions behind each result



Probability distributions behind selected headline results: the subsampling distribution of the estimate against the permutation null, with the fitted normal. Separation between the two is the significance, shown rather than asserted.



The correlation with the DJF index for every day of the water year, against the snowpack climatology, showing when in the season the signal acts.

METHOD

How this was measured

A powder day is six inches of snow

A fixed water threshold is a maritime yardstick. Cascade new snow runs about 12 % density and Colorado's about 7 %, so 25 mm of water is roughly eight inches of Cascade snow but fourteen inches of Colorado snow — counting water alone makes the driest, lightest snowpacks look storm-free. Each station's own new-snow ratio is measured from the overlap of its depth and water records, then applied to its much longer water record. Across 508 stations the median ratio is 7.5 inches of snow per inch of water.

Regions are ranges, not states

A state boundary is not a snow boundary. Tahoe spans California and Nevada, the Bitterroots and Selkirks span Montana and Idaho, the Tetons span Wyoming and Idaho, the Sangre de Cristo spans Colorado and New Mexico. Each region is anchored on a mountain range around real ski terrain and labelled with its snow climate.

Stations are weighted by elevation

A gauge represents a ski area only insofar as it shares its snow climate, and elevation is most of that. Each region carries its served base-to-summit band; stations are weighted by a Gaussian on the vertical gap to that band and another on horizontal distance, so a valley gauge below the lifts does not speak for a mountain.

Significance, and the correction for asking often

The modified bootstrap of Brown & Harper (2026): omit a random 20 % of winters, refit, repeat ten thousand times, and require the 2σ bounds of the resulting near-Gaussian distribution to exclude zero. That raw rule fires on about 30 % of pure noise, because the subsample spread understates the sampling error by the delete-*d* factor $\sqrt{((n-d)/d)} = 2$, so a calibrated version and a permutation p-value are reported alongside. Benjamini-Hochberg control is then applied across the entire grid.

Two views of the same data

The statistical tests run on per-station detrended z-scores, so a long-term warming trend cannot pose as an ENSO signal. The physical composites are raw averages by phase – what actually happened. The two are matched explicitly on region, window and measure rather than assumed to line up.

Windows that carry a season

Early season (1 Nov – 15 Dec), Holidays (16 Dec – 5 Jan), Midwinter (6 Jan – 28 Feb) and Spring (1 Mar – 15 Apr). They abut without gaps. These replace the hydrologic periods used in water-supply work, which are built around runoff timing rather than revenue.

COVERAGE

Regions and station coverage

Effective station count is Kish's measure applied to the elevation weights. Regions with few stations inside the served band – notably southern Utah and the Snowy Range – should be read with more caution, because SNOTEL does not reach those elevations locally.

SKI REGION	STATES	CLIMATE	SERVED BAND (FT)	STATIONS	EFFECTIVE	WEIGHTED ELEV (FT)	IN BAND
Beartooth & Absaroka (Red Lodge)	MT+WY	continental	7,016–9,416	16	14.4	8,143	11
Bighorn Mtns	WY	continental	7,500–9,500	15	13.8	8,747	12
Bitterroot & Lolo (Montana Snowbowl)	MT+ID	transitional	5,000–7,600	13	10.9	5,978	10
Black Hills (Terry Peak)	SD	continental	5,800–7,064	3	2.9	6,358	3
Blues & Wallowas (Anthony Lakes)	OR	transitional	7,100–8,000	18	13.7	5,834	2
Bridger/Gallatin/Madison (Big Sky)	MT	continental	6,400–10,000	20	17.2	7,848	20
Central Cascades (Stevens/Snoqualmie)	WA	maritime	3,000–5,845	20	18.1	3,758	17
Central Oregon (Mt Bachelor)	OR	transitional	6,300–9,065	10	7.3	5,220	0
Chugach (Alyeska)	AK	maritime	250–2,750	11	9.5	1,413	7
E Cascades rain shadow (Mission Ridge)	WA	transitional	4,570–6,820	7	5.9	4,744	2
Elk Mtns (Aspen/Crested Butte)	CO	continental	7,945–12,162	14	12.8	10,172	14
Front Range (Winter Park/Loveland/A-Basin)	CO	continental	9,000–13,050	27	23.6	10,314	23
Gore & Tenmile (Vail/Summit/Copper)	CO	continental	8,120–12,998	24	21.8	10,417	24
Jemez & N New Mexico	NM	continental	8,500–11,500	9	7.5	9,473	7
Mt Hood	OR	maritime	4,500–8,540	11	9.4	4,258	2
N Sierra / Tahoe (Palisades/Heavenly/Rose)	CA+NV	maritime	6,200–10,067	27	25.5	7,599	26
North Cascades (Mt Baker)	WA	maritime	3,500–5,089	15	11.8	4,227	9
Park Range (Steamboat)	CO	continental	6,900–10,568	18	15.9	9,367	16
Ruby Mtns & NE Nevada	NV	continental	6,500–10,000	11	10.0	7,866	10

SKI REGION	STATES	CLIMATE	SERVED BAND (FT)	STATIONS	EFFECTIVE	WEIGHTED ELEV (FT)	IN BAND
S Sierra (Mammoth/June)	CA	maritime	7,953–11,053	2	1.7	9,038	2
S Utah (Brian Head/Eagle Point)	UT	continental	9,600–10,970	30	25.0	9,396	11
S Washington Cascades (Crystal/White Pass)	WA	maritime	4,400–7,012	24	19.0	4,668	12
Sacramento Mtns (Ski Apache)	NM	continental	9,600–11,500	1	1.0	10,250	1
San Francisco Peaks (AZ Snowbowl)	AZ	continental	9,200–11,500	8	4.8	8,429	1
San Juans (Telluride/Purgatory/Wolf Ck)	CO	continental	8,725–13,150	27	25.5	10,495	27
Sangre de Cristo (Taos/Santa Fe)	NM+CO	continental	9,200–12,481	20	17.1	10,114	17
Sawtooth & Smoky (Sun Valley)	ID	intermountain	5,750–9,150	20	18.1	7,569	18
Selkirk & Cabinet (Schweitzer/Silver)	ID+MT	transitional	4,000–6,400	15	13.3	5,206	14
Snowy Range & Sierra Madre	WY	continental	8,798–9,663	21	19.1	9,324	6
Spring Mtns (Lee Canyon)	NV	continental	8,510–11,290	4	4.0	8,723	3
Tetons (Jackson/Targhee)	WY+ID	intermountain	6,300–10,450	15	12.3	7,943	15
Uinta Mtns	UT	continental	8,000–11,000	30	27.9	9,345	26
W Central Idaho (Brundage/Tamarack)	ID	intermountain	5,840–7,640	11	9.5	5,962	6
Wasatch (Alta/Snowbird/Park City)	UT	intermountain	6,800–11,000	31	27.8	8,173	29
White Mtns AZ (Sunrise Park)	AZ	continental	9,200–11,100	9	8.0	8,668	2
Whitefish & Flathead Range	MT	transitional	4,464–6,817	12	10.9	5,442	10
Yellowstone & Wind River	WY	continental	7,000–10,000	34	29.6	8,683	30

LIMITS

What this does not establish

- **An r^2 of 15–35 % is real and it is not a forecast.** ENSO shifts the odds for a season. Most of the year-to-year variance remains weather.
- **Snow depth records begin around 1993.** New-snow ratios are calibrated on that period and applied backward, which assumes density has been stable. Depth-based composites rest on roughly 25 winters against 45 to 60 for the water-based ones.
- **A SNOTEL gauge is not a ski area.** Even with elevation weighting these are nearby mountain stations, and they say nothing about aspect, grooming or snowmaking.
- **Overlapping regions are not fully independent** where their radii intersect, so the sign tests read slightly optimistically.
- **Six inches is a choice.** Regions with few powder days in absolute terms show large percentage swings on small differences.
- **The strength null is bounded by power.** Within-phase tests had about sixteen winters and could only have detected a correlation above roughly 0.65. The claim is that across 2158 chances nothing rose above noise, and the better-powered nested test also found nothing.

Sources – NOAA Climate Prediction Center, Oceanic Niño Index. USDA NRCS Air & Water Database, SNOTEL daily snow water equivalent, snow depth, precipitation and temperature. NOAA NCEI nClimDiv. Method after J. Brown and J. Harper (2026), *Historical evolution of snowpack capacity to buffer rain-on-snow runoff in a large Columbia River headwaters basin*, Hydrol. Earth Syst. Sci. 30, 5735–5748. False-discovery control after Benjamini & Hochberg; field significance after Wilks (2016).

Analysis code and every table in this report are reproducible from github.com/yetiskier/enso-snowpack.